Closed Native Verification Systems (CNVS): Complete Formal Theory and Emergent Security Theorem

Abstract

This paper introduces Closed Native Verification Systems (CNVS), a formal class of verification systems in which state membership is verification-dependent rather than primitive. The theory combines a typed structural universe, recursive type-preserving decomposition, randomized assignment of non-uniform fragments, restricted verifier knowledge, and global invariant checking. A conditional probabilistic theorem bounds unauthorized reconstruction by a binomial tail and yields exponential suppression when the reconstruction threshold exceeds the expected number of useful compromised fragments. An asymptotic scaling theorem shows that security can increase with system expansion under explicit assumptions.

1. Primitive Domains and Typed Universe

Let $B = \{0,1\}$ and B^* be the set of all finite binary strings. Define the primitive domains:
$Id \subseteq B^*$ (identifiers),
$D \subseteq B^*$ (declared data),
$At \subseteq B^*$ (attribute classes),
$Val \subseteq B^*$ (externally reported values),
P (finite set of logical properties),
Tau (finite set of structural types).
The structural universe is:
$U = Id \times D \times At \times Val \times Pow(P)$.
Each element $s \in U$ has the form $s = (id, d, a, v, p)$.

2. State Space

A system state at time t is:

$S(t) = (E(t), R(t), C, V\_L)$,

where $E(t) \subseteq U$ *is the set of admitted elements,* $R(t) \subseteq E(t) \times E(t)$ *is a directed relation set,* $C = \{c1, \ldots, cm\}$ *is a finite family of invariant predicates, and V_L is the local verification function.*

3. Local Verification

The local verification function is:

$V\_L : Id \times D \times At \times Val \to \{0,1\} \times Pow(P)$.

For $x = (id, d, a, v)$*, let:*

$V\_L(x) = (b, p)$,

where

$b = Form(id,d,a,v) \wedge Conv\_a(d,v)$.

Local admissibility is:

$Adm\_L(s,t) = 1 \Leftrightarrow \pi 1(V\_L(s)) = 1$.

4. Recursive Decomposition and Reconstruction

The decomposition operator is:

$D\_t : U \to Pow\_fin(U)$.

For every admissible element s:

$D\_t(s) = \{F1, \ldots, Fk\}, 1 \leq k < \infty$.

Type preservation:

$\forall s' \in D\_t(s), type(s') = type(s)$.

Reconstruction:

$Rec\_t(D\_t(s)) = s$

up to canonical encoding.

5. Non-Uniform Fragmentation

There exist fragments Fa and Fb such that:

$|I(Fa)| \neq |I(Fb)|$.

Hence decomposition is intentionally non-uniform. Some fragments contain more information than others.

6. Knowledge Restriction

Let $K\_x(t)$ denote the knowledge accessible to actor x at time t. For every external verifier v_j:
$K\_V(j,t) \subsetneq I\_G(t)$,
$K\_V(j,t) \subsetneq I\_N(i,t)$.
No verifier possesses enough information to reconstruct the entire global state.

7. Information Density

For a fragment F_j:
$\rho\_N(j,i,t) = |I\_F(j,t)| / |I\_N(i,t)|$,
$\rho\_G(j,t) = |I\_F(j,t)| / |I\_G(t)|$,
$\rho\_C(j,i,t) = \rho\_N(j,i,t)^\alpha \rho\_G(j,t)^\beta$,
where $\alpha,\beta \in [0,1]$ and $\alpha + \beta = 1$.

8. Random Assignment

Terminal fragments are assigned to external verifiers by:
$A\_t : T(t) \times \Xi \to V\_ext(t)$,
where Ξ is a probability space and $T(t)$ is the set of terminal fragments.

9. Axiom I: Verification-Dependent Existence

For every element s and time t:
$s \in E(t) \Rightarrow Adm\_L(s,t) = 1$.
Existence inside the system is a derived consequence of successful verification.

10. Axiom II: Randomized Non-Uniform Fragmentation

Every admissible element is recursively decomposed into non-uniform fragments that are assigned randomly:
$D\_t(s) = \{F1, \ldots, Fk\}$,
$\exists Fa,Fb : |I(Fa)| \neq |I(Fb)|$,
$A\_t : T(t) \times \Xi \rightarrow V\_ext(t)$,
$Rec\_t(D\_t(s)) = s$.

11. Axiom III: Non-Reducibility of Global Validity

Global validity is not reducible to local validity:
$V\_G(S) \neq \wedge\_\{s \in E\}\ V\_L(s)$.
Instead:
$V\_G(S) = LocalOK(E) \wedge Cons\_R(R,E) \wedge Inv\_C(S)$.

12. Fundamental Lemmas

Lemma 1. Local admissibility is necessary for state membership.
Lemma 2. Local validity is insufficient for global validity.
Lemma 3. Type is preserved across decomposition depth.
Lemma 4. Reconstruction conserves payload information.
Lemma 5. Metadata overhead permits encoded expansion.
Lemma 6. Average fragment density decreases under controlled subdivision.

13. Principle of Knowledge Restriction

As recursive decomposition increases:
$|K\_V(j,t)| / |I\_G(t)| \rightarrow 0$.
Therefore:
More fragmentation ⇒ Less local knowledge ⇒ Greater distributed ignorance ⇒ Greater security.

14. Probabilistic Security Model

Let:
k = number of terminal fragments,
r = reconstruction threshold,
C = colluding verifier count,
q = total verifier count,
$p = C/q$,
λ_j = probability that a compromised fragment is useful,
$p_{eff} = p\, \lambda_{max}$.
Define $X_j = 1$ if fragment j is compromised and useful, and
$X = \Sigma_j X_j$.
Unauthorized reconstruction occurs when:
$Rec^* = \{X \geq r\}$.

15. Conditional Security Theorem

If X is stochastically dominated by $Bin(k, p_{eff})$, then:
$Pr(Rec^*) \leq \Sigma_{i=r}^{k} \binom{k}{i} p_{eff}^i (1-p_{eff})^{(k-i)}$.

Proof.
By definition, unauthorized reconstruction requires at least r useful compromised fragments.
Under stochastic domination, $X \leq_{st} Y$ where $Y \sim Bin(k,p_{eff})$. Therefore:
$Pr(X \geq r) \leq Pr(Y \geq r)$,
which equals the stated binomial upper tail. QED.

16. Exponential Chernoff Bound

Let $\mu = k\, p_{eff}$. If $r > \mu$, then:
$Pr(Rec^*) \leq \exp(-(r-\mu)^2 / (2\mu + (r-\mu)))$.

Proof.
Apply the multiplicative Chernoff bound to $Y \sim Bin(k,p_{eff})$. QED.

17. Emergent Security Scaling Theorem

Let n denote system scale. Assume:
$k(n) \to \infty$,
$p\_eff(n)$ remains bounded or decreases,
$r(n) - k(n)p\_eff(n) \to +\infty$.
Then:
$Pr(Rec^*\_n) \to 0$.

Proof.
Substitute the scaling assumptions into the Chernoff exponent. The exponent diverges to $+\infty$, hence the probability tends to zero. QED.

18. Interpretation

Under the stated assumptions, system growth increases fragmentation and epistemic restriction. As a result, unauthorized reconstruction becomes exponentially less probable. Security is therefore an emergent property of scale.

19. Scientific Positioning

CNVS combines concepts from information theory, probability theory, cryptography, formal methods, graph theory, and epistemic logic into a unified mathematical framework.

20. Open Problems

1. Formal leakage models.
2. Dynamic collusion models.
3. Cryptographic reductions.
4. Information-theoretic lower bounds.
5. Empirical validation.

Appendix A. Minimal Formal Definition

A structure

M = (U, E, R, C, D, A, V_L, V_G, Rec, K)

is a Closed Native Verification System if and only if:

1. U is a typed finite-binary universe.

2. Membership requires local admissibility.

3. Global validity requires relation consistency and invariants.

4. Decomposition is finite, recursive, and reconstructible.

5. Verifier knowledge is strictly restricted.

6. Terminal assignments are randomized.

7. Invalid elements and states are rejected.

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